

CURRICULUM VITAE – ELIO MOREAU

PERSONAL INFORMATION

Elio Moreau
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EXPERIENCES

- Research internship** Capital Fund Management, Paris *September 2025-March 2026*
Research for the Macro Alpha team to create a method to detect and characterize the multi-modality in data from unknown distributions using methods and concepts of machine learning and generative modeling such as stochastic interpolants and reaction paths. Also implementation of the method in pytorch.
- M2 research internship** MPQ, Paris Cité University, Paris *April 2025-July 2025*
Formulated a new theoretical model to describe and predict alterations in chemical reaction kinetics within optical cavities which explicitly accounts for collective effects in polaritonic chemistry
- Math Intelligence Engineer** Mercor, Meta, Remote, Part Time *February 2025-August 2025*
Evaluating the performance of a cutting-edge AI model in collaboration with the AI researchers responsible on its development.
- M1 research internship** ENS-PSL, Paris *February 2024-July 2024*
Analytical calculation of transport properties of SYK models, which are quantum many-body systems with random variable couplings, first in states out of thermal equilibrium and then in for systems with the number of quantum degrees of freedom increases with time.
- Atomic Force MACROscope project** ENS-PSL, Paris *May 2023*
Building an Atomic Force Microscope based on a centimetric harmonic oscillator that will have a nanometric resolution.
- Research Assistant** Micromégas lab, ENS-PSL, Paris *February 2023-June 2023*
Experimental study of ionic current rectifications in slits and channels having width of tens of nanometers. Using these properties to create memristors and designing memristive circuits.
- EPFL Excellence Research Internship Program** EPFL, Lausanne *July 2022-September 2022*
Created an interactive user interface for data captured with adaptive microscopy techniques that also allowed on the fly data filtering, segmentation and image analysis.

PAPERS

- ICML 2026 Probing the Geometry of Diffusion Models with the String Method**
Elio Moreau, Florentin Coeurdoux, Grégoire Ferré, Eric Vanden-Eijnden.
International Conference on Machine Learning (ICML), 2026.

EDUCATION

- ENS-PSL, ENS Diploma** Paris *September 2023 - Expected June 2026*
Double Major in Physics and Mathematics
- ENS-PSL, ICFP Master in Theoretical Physics** Paris *September 2023 - July 2025*
GPA : 15.835/20
Classes on : Data Driven Modelling, Advanced Statistical Physics, QFT, DMFT, DFT, Machine Learning, Condensed Matter Physics, Lie Algebras, Statistical Field Theory, General Relativity...
- Paris-Saclay University, Master in mathematics** Paris *September 2026 - Expected June 2026*
Classes on : Probability, Statistics, Stochastic Processes, Algebra, Topology, Differential Geometry, Theoretical Neuroscience, Stochastic Differential Equations...
- EPFL, Bachelor in Physics** Lausanne *September 2020 - July 2023*
Bachelor's GPA : 5.76/6
Best GPA among Physics Bachelors in 2023, and third best among all EPFL Bachelors in 2023
Classes on : Calculus, Linear Algebra, Statistics, Classical Physics, Quantum Physics, Statistical Physics, Object Oriented Programming...
- Liceo Scientifico Pacinotti, High School Diploma** Cagliari *September 2015 - July 2020*
Final grade : 100 cum laude
Did a one year cultural exchange with **Higashi Hagoromo High School** at Takaishi, in Japan

ASSOCIATIONS	Délégation Générale ENS, Paris	<i>May 2023-October 2024</i>
	The Délégation Générale is a collegial body composed of four elected students which is in charge of the distribution of rooms in the dormitory, the allocation of places in the car parks, the distribution of trunks and also acts as a mediator between the students and the administrative services.	
	EPFL Xplore project EPFL, Lausanne	<i>September 2021-September 2022</i>
	Designed and built the science bay of the rover that represented EPFL to the European Rover Challenge in Poland and got ranked second. Use of Fusion360 to design the components.	
	EPFL Space Race Project EPFL, Lausanne	<i>October 2020-May 2021</i>
	Design and construction of the parachute and recovery system of a rocket meeting the requirements of the official Level 2 certification of the Tripoli Amateur Launchers Association, The objective, successfully reached, was to propel a raw egg up to 1km in height and return it intact to the ground. Used CATIA to design the components	
AWARDS	• Bronze Medal, European Physics Olympiad, 2020 • Bronze Medal, International Earth Sciences Olympiad, 2017 • Silver Medal, Italian National Math Olympiad, 2020	
	ENS-PSL International Selection	<i>2023-2026</i>
	Scholarship given by ENS - PSL to 10 students in sciences among more than 1500 applicants	
	National Register for Excelling Students	<i>2017,2108,2020</i>
	Scholarship granted for three years (2017, 2018, 2020), having obtained outstanding results in Olympiads.	
TEACHING	Teacher Private classes, Paris	<i>September 2023 - April 2024</i>
	Gave private lessons for high-school and bachelors students in maths, physics, chemistry and computer sciences.	
	Teaching Assistant AdmiUniv, Paris	<i>September 2023 - April 2024</i>
	Gave mock oral exams in math and physics to students preparing admission contests for prestigious higher education establishments in engineering	
	Teaching Assistant EPFL, Lausanne	<i>September 2021 - July 2022</i>
	Teaching assistant in the courses : Advanced Analysis I for Physics, in the General Physics I : Mechanics for Biomedical Engineering, General Physics I : Mechanics english option, Analysis 2 for Environment engineering	
	Mentor EPFL, Lausanne	<i>September 2021 - July 2022</i>
	Mentor in the EPFL Mentoring Program for first year students	
PROGRAMMING SKILLS	Programming languages : C++, Python, Qt.	
	Office softwares : Microsoft Office, Open Office, LaTeX.	
	Scientific softwares : Matlab, Mathematica, LabVIEW.	
	Design softwares : Fusion360	
	Machine learning : PyTorch, scikit-learn	
LANGUAGES	Italian - native proficiency (ILR 5)	
	French - native proficiency (ILR 5)	
	English - full working proficiency (ILR 4, certified C1)	
	Japanese - limited working proficiency (ILR 2, certified JLPT N3)	